ONE-PAGE SUMMARY · I3X NEXUS PROFILE, DRAFT 0.4, WITH A WORKING REFERENCE IMPLEMENTATION

i3X Nexus: one governed, redundant endpoint per plant

The i3X Nexus is one endpoint per plant site that serves live values, history, business records and media (video, images, audio) through CESMII's open i3X 1.0 interface. Every change it makes to a plant or business system is **authorised, recorded, reversible and checked afterwards**, and it keeps running through the loss of a server, its database primary, or the site's internet link.

> **Why it matters.** OPC UA, i3X and MCP already standardise how plant data moves and how AI models reach tools. None of them says who may *change* something, under whose authority, with what record of why, how it is undone, or how anyone checks it worked. AI agents now act on plant data, so that gap matters now. The nexus fills it inside the existing i3X contract: an i3X client that has never heard of it still works.

How a governed change works

- A person or an AI agent submits a **write request**, never a direct write. The request carries who asked, the decision that justifies it, and how to undo it.
- The site's **authority limits** decide who must approve, by exposure, materiality and safety triggers. Agents may ask but never approve, and nobody approves their own request.
- Before applying, the nexus checks that the target has not changed since the request. It then writes conditionally and **reads the result back**. A mismatch is undone automatically.
- Every step is a hash-chained record, so the audit trail can't be edited unnoticed. Rollback is a new, approved, linked request; nothing is ever erased.
- A falsification condition on the decision lets a verifier report later whether the change did what it claimed.

What it connects

- **Live data:** OPC UA, MQTT and Sparkplug B. A lost or stale source shows as Bad or Uncertain, never as a fresh-looking number.
- **History:** the plant's own historian (Ignition SQL on Postgres, MySQL or SQL Server), with no copy kept.
- **Systems of record:** work orders and other records (CMMS, MES), shown read-only and changed only through write requests.
- **Decisions and authority:** decision records, outcome observations and authority limits, so an operations platform that holds decision rights can govern what agents and systems do.
- **Media:** live camera and audio through a gateway, with signed short-lived viewer links. An alarm cuts a clip that is linked to it.

WRITE TIER	WHAT	GATE
`W0 publish`	A system writes into its own namespace: decisions, observations, signals	Namespace ownership, checked by the server
`W1 record`	Change a record in a system of record: open a work order, place a quality hold	Authority limit and approval, conditional apply, read-back
`W2 setpoint`	Change a value the process acts on	Hard limits on range and step, approval by a named person every time, read-back, paused if redundancy is degraded

Built to survive failures and load

- **Two nodes** serve reads. One node, the governor, makes writes; if it dies, the other takes over and finishes in-flight writes exactly once.
- **A redundant database** with synchronous replication and a three-member quorum, so no acknowledged write is lost.
- **Each site runs on its own** when its internet link is down. A central enterprise copy shows the site's values as Uncertain, then catches up alarms first.
- **Priority lanes:** alarms, approvals and rollbacks are never dropped under load. Bulk history is turned away first, with a clear "retry later".
- **Security:** OAuth 2 sign-in or client certificates, HTTPS, one identity per person or agent, and a design that follows IEC 62443.

Proven, not just specified

CESMII i3X 1.0 suite	Full Compliance, 0 failures, 0 warnings over HTTPS
Conformance during failover	Passes with the write node killed mid-run
20 forced node kills	Writes resume within 12.5 s (target 30 s); 0 lost, 0 duplicated
Database primary killed	Writes resume within 20 s, both directions; audit chain intact
Setpoint fuzzing	10,000 values tried; none outside the limits accepted
Live lab	Running on two lab servers, with a third as tie-breaker

Open, vendor-neutral profile of CESMII i3X 1.0, offered for comment and submission to CESMII. Status: draft 0.4. The reference implementation meets conformance level 4 of 4.

Synthetic data only.